Buoyancy & Archimedes' Principle

Buoyant Force

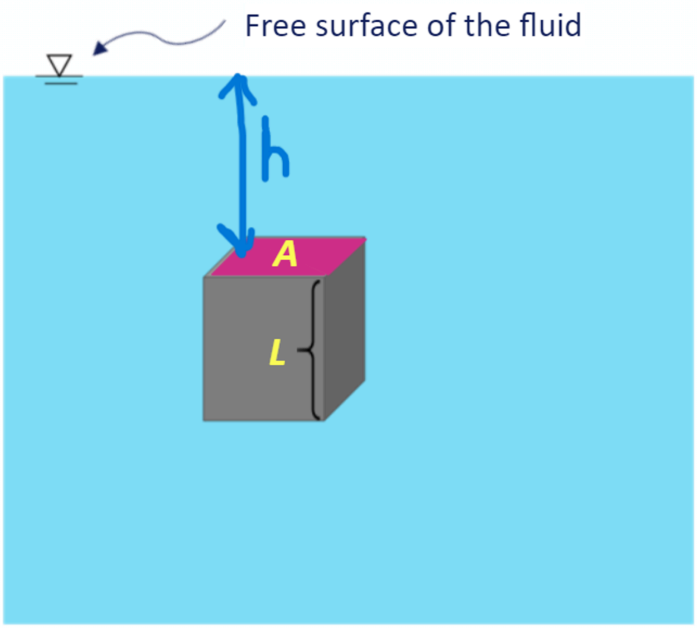
The buoyant force is the upward force exerted by a fluid on an object.

The formula is as so:

$$F_b = \rho g V_{\text{displaced}}$$

▼ Derivation

Consider a block suspended in a fluid.



The block has a base area of A, a height of L, and has a depth of h from the surface of the fluid to top surface

The fluid will then exert pressure on all sides of the block not just the cube, however not equally.

For pressure exerted on the sides, the pressure exerted on the left will be cancelled out by the pressure on the right, so we can ignore them. (similar for front and back)

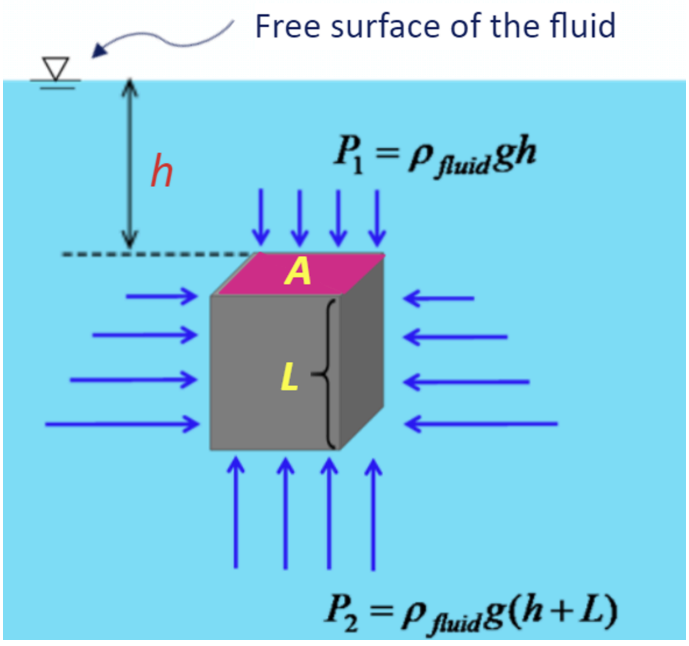
On the top of the box, the (downward) force due to the hydrostatic pressure is

$$F_{top} = P_{top}A = \rho g h A$$

On the bottom of the box, the (upward) force due to the hydrostatic pressure is

$$F_{bottom} = P_{bottom}A = \rho g(h + L)A$$

Notice how there will be more force, as it is at a greater depth

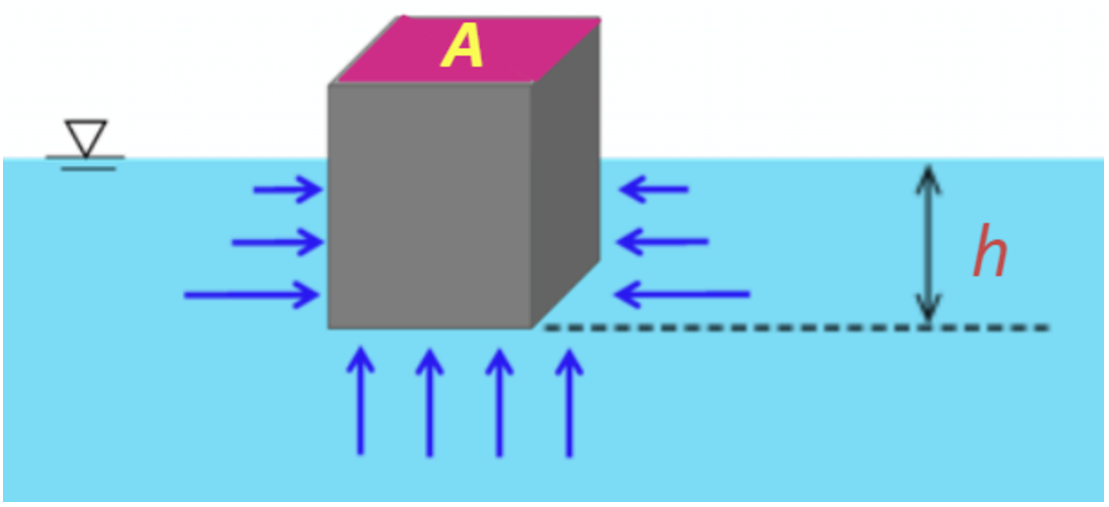


When we consider these two forces, we notice there is a net force upwards (buoyant force), which we can find as:

$$\begin{aligned} F_B &= F_{bottom} - F_{top} \\ &= \rho g(h + L)A - \rho g h A \\ &= \rho g A(h + L - h) \\ &= \rho g A L \\ &= \rho g V \end{aligned}$$

Notice how AL is just the volume of the block, thus giving us our equation.

For objects that float, a similar derivation and formula holds.



The upper portion of the block has no hydrostatic pressure exerting on it, thus the buoyant force is only due to the pressure at the bottom. (aka equal)

The force at the bottom due to the pressure is:

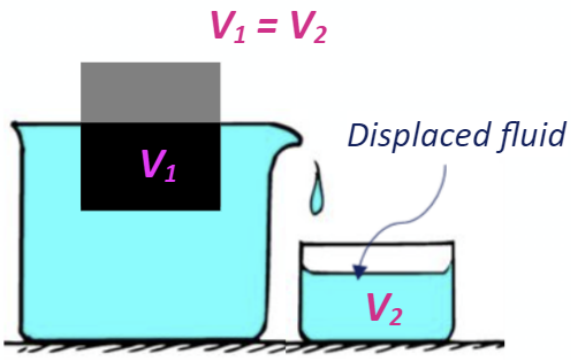
$$F_B = \rho g h A$$
$$= \rho g V_{\text{displaced}}$$

Note that V is no longer the volume of the whole block, but only the submerged portion, or the volume the cube displaces out of the water.

Some things to note about this formula is that
It is not dependant on weight or any external

Archimedes' Principle

Consider the setup shown:



When the block is placed into the fluid, it will place a certain amount of fluid of volume V1. It then displaces an equal volume V2 of fluid out of the vessel.

Archimedes' Principle is that the magnitude of the buoyant force is equal to the weight of the displaced fluid, i.e.

$$F_B = W_{\text{displaced}}$$

We can derive this as so:

$$\begin{aligned} F_B &= \rho g V_{\text{displaced}} \\ &= \frac{m}{V} g V \\ &= mg \\ &= W_{\text{displaced}} \end{aligned}$$

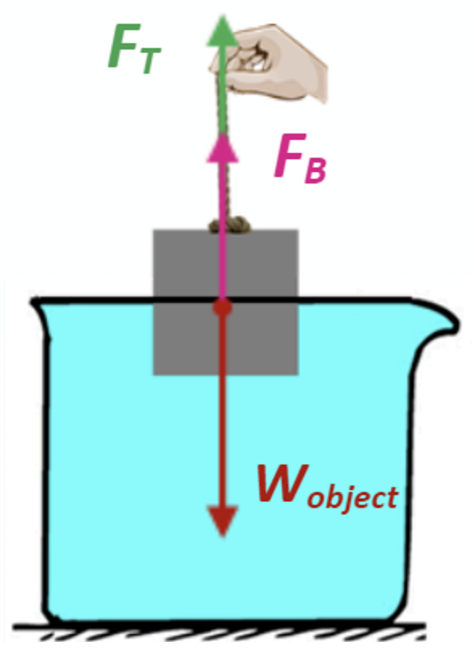
Free Body Diagrams w/ Buoyancy

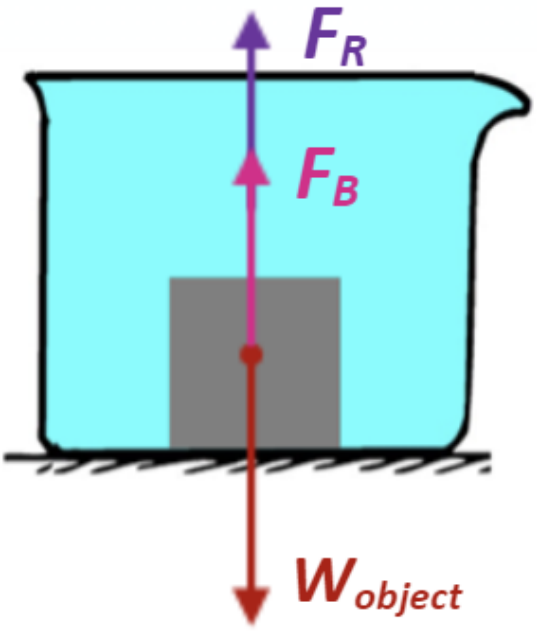
Similar to frictional forces in Mechanics, the buoyancy force is another force that acts on an object when partially/fully submerged in a fluid, like weight and normal reactions.

We can draw buoyancy on the free body diagram of a submerged object, and can use equilibrium concepts to solve for weight/other forces.

One thing to note is that for floating blocks, the object is at equilibrium, meaning $W = F_B$, not $W < F_B$ because when we consider $W < F_B$, the object will accelerate upwards (net force is upwards), however a floating object does not accelerate.

Some examples considering tension or normal forces:


F_T
F_B
W_{object}


F_R
F_B
W_{object}